Unit 2
First Order Linear and Non-linear Differential Equation

Linear First Order Differential Equation

A differential equation of the form

$$\alpha(x)\frac{dy}{dx}+\beta(x)y=\gamma(x) \quad \ldots \quad (1)$$

is said to be a first order linear differential equation in general form.

If $\alpha(x)\neq 0$, dividing both sides of (1), it reduces to

$$\frac{dy}{dx}+p(x)y=q(x) \quad \ldots \quad (2)$$

where $p(x)$ and $q(x)$ are functions of x or constant, which is known as first-order linear differential equation in standard form.

A first order differential equation that cannot be put in the form (1) or (2) is called non-linear.

The equation (2) is called homogeneous first order linear differential equation if $\frac{dy}{dx}+p(x)y$ which has $y=0$ as a solution. This is called the trivial solution of $y'+py=0$. All other solutions are called non-trivial.

Singular Points of (1)

The value of x for which $\alpha(x)=0$ are called the singular points of the equation (1).

Example Solve $(4+t^2)\frac{dy}{dx}+2ty=4t$

Solution

The given differential equation is

$$(4+t^2)\frac{dy}{dx}+2ty=4t \quad \ldots \quad (1)$$

which is linear first order differential equation. Since LHS of (1) is the derivative of the product $(4+t^2)y$, it can be written as

$$\frac{d}{dt}[(4+t^2)y]=4t$$

Integrating it we get

$$(4+t^2)y=\int 4t\,dt$$

$$\Rightarrow (4+t^2)y=2t^2+c$$

$$\Rightarrow y=\frac{2t^2+c}{4+t^2}.$$

This is the required general solution of (1).

Integrating Factor

All linear first-order differential equations do not have LHS which can be put as a single differential as in the above example. We can find a function $\mu(x)$, which when multiplies the equation

$$\frac{dy}{dx}+p(x)y=q(x),$$

the left hand side of this becomes perfect differential. Such a function $\mu(x)$ is called an integrating factor.

Thus, an integrating factor (I.F.) of a first order linear differnretial equation

$$\frac{dy}{dx}+p(x)y=q(x) \quad \ldots \quad (1)$$

is a function $\mu(x)$ which when multiples (1) makes its LHS a perfect differential.

To Find the Integrating Factor

The linear first order differential equation is

$$\frac{dy}{dx} + p(x)y = q(x) \quad \dots \quad (1)$$

where $p(x)$ and $q(x)$ are functions of x or constants.

Let $\mu(x)$ be the intergrating factor (I.F.) of (1). Then, when (1) is multipliced by $\mu(x)$, we get

$$\mu(x)\frac{dy}{dx} + p(x)\mu(x)y = q(x)\mu(x) \quad \dots \quad (2)$$

Since $\mu(x)$ is an I.F. the LHS of (2) is the derivative of the product $\mu(x)\, y$ such that

$$\frac{d}{dx}\mu(x) \;=\; p(x)\mu(x).$$

That is,

$$\mu(x)\frac{dy}{dx} + p(x)\mu(x)y = \frac{d}{dx}\,[\mu(x).y] = \mu(x)\frac{dy}{dx} + \frac{d\mu(x)}{dx}y.$$

Therefore,

$$\frac{d\mu(x)}{dx} = p(x)\mu(x).$$

Now,

$$\frac{d\mu(x)}{dx} = p(x)\mu(x)$$

$$\Longrightarrow \frac{d\mu(x)}{\mu(x)} = p(x)dx$$

Integrating, we get

$$\ln|\mu(x)| = \int p(x)\,dx + c$$

$$\Longrightarrow \mu(x) = e^{\int p(x)dx+c}$$

In particular, taking $c = 0$, we have

$$|\mu(x)| = e^{\int p(x)dx}$$

Since $e^{\int p(x)dx} > 0$ for all x we have

$$\mu(x) = e^{\int p(x)dx}$$

Hence, the integrating factor of $\frac{dy}{dx} + p(x)y = q(x)$ is

$$I.F. = e^{\int p(x)dx}$$

Example Solve $\frac{dy}{dx} - 2y = x^2e^{2x}$

Solution

The given differential equation is

$$\frac{dy}{dx} - 2y = x^2e^{2x} \quad \dots \quad (1)$$

where, $p(x) = -2, q(x) = x^2e^{2x}$

$$\therefore I.F. = e^{\int p(x)dx} = e^{\int -2dx} = e^{-2x}.$$

Multiplying both sides of (1) by e^{-2x}, we get

$$e^{-2x}\frac{dy}{dx} - 2e^{-2x}y = x^2$$

$$\Longrightarrow \frac{d}{dx}(e^{-2x}.y) = x^2$$

Integration implies

$$e^{-2x}.y = \frac{x^3}{3} + c \Rightarrow y = \frac{x^3 e^{2x}}{3} + ce^{2x}$$

Examples

1. Find the general solution of the given differential equation, and use it to determine the behaviour of solutions as $t \to \infty$.

$$(i) \quad y' + 3y = t + e^{-2t}$$

Solution

The given equation is

$$y' + 3y = t + e^{-2t} \qquad \ldots \qquad (1)$$

Here, $P = 3, Q = t + e^{-2t}$.

$$\therefore \; I.F. = e^{\int P dt} = e^{\int 3dt} = e^{3t}$$

Multiplying (1) by e^{3t}, we get

$$e^{3t}\frac{dy}{dt} + 3e^{3t}y = te^{3t} + e^t$$

$$\Rightarrow \frac{d}{dt}(y.e^{3t}) = te^{3t} + e^t$$

Integrating, we get

$$y.e^{3t} = \int (te^{3t} + e^t)dt$$

$$\Rightarrow y.e^{3t} = \int te^{3t}\,dt + \int e^t\,dt$$

$$= t\int e^{3t}\,dt - \int\left(\frac{dt}{dt} \times \int e^{3t}\,dt\right)dt + e^t \; Integrating\ by\ parts$$

$$= t\frac{e^{3t}}{3} - \int 1.\frac{e^{3t}}{3}dt + e^t$$

$$= t\frac{e^{3t}}{3} - \frac{e^{3t}}{9} + e^t + c$$

where, c is an arbitrary constant of integration.

Hence, the required general solution of (1) is

$$y = \frac{t}{3} - \frac{1}{9} + e^{-2t} + ce^{-3t}$$

As $t \to \infty$, the terms e^{-2t} and ce^{-3t} both varish i.e. $\lim_{t\to\infty} e^{-2t} = 0$

And $\lim_{t\to\infty} ce^{-3t} = 0$, so the general solutions coverage to the function

$$y = \frac{t}{3} - \frac{1}{9}$$

Tha is, the solution curves are aspmptotic to

$$y = \frac{t}{3} - \frac{1}{9} \; as\ t \to \infty.$$

$ii. \quad y' - 2y = t^2 e^{2t}$

$$I.F. = e^{-2t}, y(t) \; = ce^{2t} + \frac{t^3 e^{2t}}{3}$$

$$y \to \infty \; and \; t \; \to \infty$$

$iii. \quad y' + y = 5sin2$

$$t,.F. = e^t, y(t) = ce^{-t} + sin2t - 2cos2t$$

$$y(t) \to sin2t - 2cos2t \; as\ t \to \infty$$

That is, y is asymptotic to $sin2t - 2cos2t$ as $t \to \infty$.

2. Find the solution of the given IVP.

$$a. \quad ty' + (t+1)y = t, y(ln2) = 1, t > 0$$

Solution

The given differential equation is

$$ty' + (t+1)y = t, y(ln2) = 1, t > 0 \quad \dots \quad (1)$$

$$\Rightarrow \frac{dy}{dt} + \left(1 + \frac{1}{t}\right) y = 1$$

Here, $P = \left(1 + \frac{1}{t}\right)$ and $Q = 1$

$$\therefore \ I.F. = e^{\int P dt} = e^{\int \left(1+\frac{1}{t}\right)dt} = e^{t+lnt} = e^t . e^{lnt} = te^t$$

Multiplying both sides of (1) by te^t, we get

$$te^t \frac{dy}{dt} + te^t \left(1 + \frac{1}{t}\right) y = te^t$$

$$\Rightarrow te^t \frac{dy}{dt} + e^t(t+1)y = te^t$$

$$\Rightarrow \frac{d}{dt}(y.te^t) = te^t$$

Integrating, we get

$$y.te^t = \int te^t \, dt$$

$$\Rightarrow y.te^t = t\int e^t \, dt - \int \left(\frac{dt}{dt}\int e^t \, dt\right) dt, integration\ by\ parts$$

$$\Rightarrow y.te^t = te^t - e^t + c$$

$$\Rightarrow y = 1 - \frac{1}{t} + \frac{c}{te^t} = \frac{te^t - e^t + c}{te^t} = \frac{t - 1 + ce^{-t}}{t}, t \neq 0$$

Therefore, the general solution is

$$y = \frac{t - 1 + ce^{-t}}{t}, t \neq 0 \quad \dots \quad (2)$$

Using the initial condition $y(ln2) = 1$, (2) gives

$$1 = \frac{ln2 - 1 + ce^{-ln2}}{ln2}$$

$$\Rightarrow ln2 = ln2 - 1 + c\frac{1}{e^{ln2}} = ln2 - 1 + c\frac{1}{2}$$

$$\Rightarrow 0 = -1 + c\frac{1}{2} \Rightarrow c = 2$$

Therefore, the spelific solution is

$$y(t) = \frac{t - 1 + 2e^{-t}}{t}, t \neq 0$$

$b.$ $t^2y' + 4t^2y = e^{-t}, y(-1) = 0, t < 0$

$c.$ $y' + \frac{2}{t}y = \frac{cost}{t}, y(\pi) = 0, t > 0.$

Example Multiply the differential equation $y' + \frac{1}{2}y = \frac{1}{2}e^{\frac{t}{3}}$ by $\mu(t)$ and determine an expression for $\mu(t)$ so as to make it exact. Hence, find its general equation. Draw the integral curves for different values of arbitrary constant. Make sure of one of the integral curves passes through (0,1). Discuss the behavior of He integral curves.

Solution

The given differential equation is

$$y' + \frac{1}{2}y = \frac{1}{2}e^{\frac{t}{3}} \quad \dots \quad (1)$$

Multiplying (1) by $\mu(t)$ we get

$$\mu(t)y' + \frac{1}{2}\mu(t)y = \frac{1}{2}\mu(t)e^{\frac{t}{3}} \quad \dots \quad (2)$$

If LHS of (2) becomes exact differential, we have

$$\mu(t)y' + \frac{1}{2}\mu(t)y = \frac{d}{dt}(\mu(t).y) = \mu(t)\frac{dy}{dt} + y\frac{d\mu(t)}{dt}$$

That is,

$$\mu(t)y' + \frac{1}{2}\mu(t)y = \mu(t)\frac{dy}{dt} + y\frac{d\mu(t)}{dt}$$

$$\Rightarrow \frac{1}{2}\mu(t) = \frac{d\mu(t)}{dt}$$

$$\therefore \frac{d\mu(t)}{dt} = \frac{1}{2}\mu(t)$$

$$\Rightarrow \frac{d\mu(t)}{\mu(t)} = \frac{1}{2}dt$$

Integration implies

$$ln|\mu(t)| = \frac{1}{2}t + ln|c|$$

$$\Rightarrow ln\left|\frac{\mu(t)}{c}\right| = \frac{1}{2}t$$

$$\Rightarrow \left|\frac{\mu(t)}{c}\right| = e^{\frac{1}{2}t}$$

$$\Rightarrow \frac{\mu(t)}{c} = e^{\frac{1}{2}t}, since\ e^{\frac{1}{2}t} > 0\ for\ all\ t$$

$$\Rightarrow \mu(t) = ce^{\frac{1}{2}t}$$

For particular taking $c = 1$, we get

$$\mu(t) = e^{\frac{t}{2}}$$

Now putting $I.F.\ \mu(t) = e^{\frac{t}{2}}$ in (2), we get

$$e^{\frac{t}{2}}\frac{dy}{dt} + \frac{1}{2}e^{\frac{t}{2}}y = \frac{1}{2}e^{\frac{t}{2}} \times e^{\frac{t}{3}}$$

$$\Rightarrow \frac{d}{dt}\left(e^{\frac{t}{2}}.y\right) = \frac{1}{2}e^{\frac{5t}{6}}$$

Integration implies

$$e^{\frac{t}{2}}.y = \frac{1}{2}\int e^{\frac{5t}{6}}\,dt$$

$$\Rightarrow e^{\frac{t}{2}}.y = \frac{1}{2} \times \frac{e^{\frac{5t}{6}}}{\frac{5}{6}} + c$$

$$\Rightarrow e^{\frac{t}{2}}.y = \frac{3}{5}e^{\frac{5t}{6}} + c$$

$$\Rightarrow y = \frac{3}{5}e^{\frac{t}{3}} + ce^{-\frac{t}{2}}$$

Therefore, the general solution is

$$y = \frac{3}{5}e^{\frac{t}{3}} + ce^{-\frac{t}{2}}$$

where, c is an arbitrary constant.

As one of the integrals passes through (0, 1), we have,

$$1 = \frac{3}{5}e^{0} + ce^{0} \Longrightarrow c = 1 - \frac{3}{5} = \frac{2}{5}$$

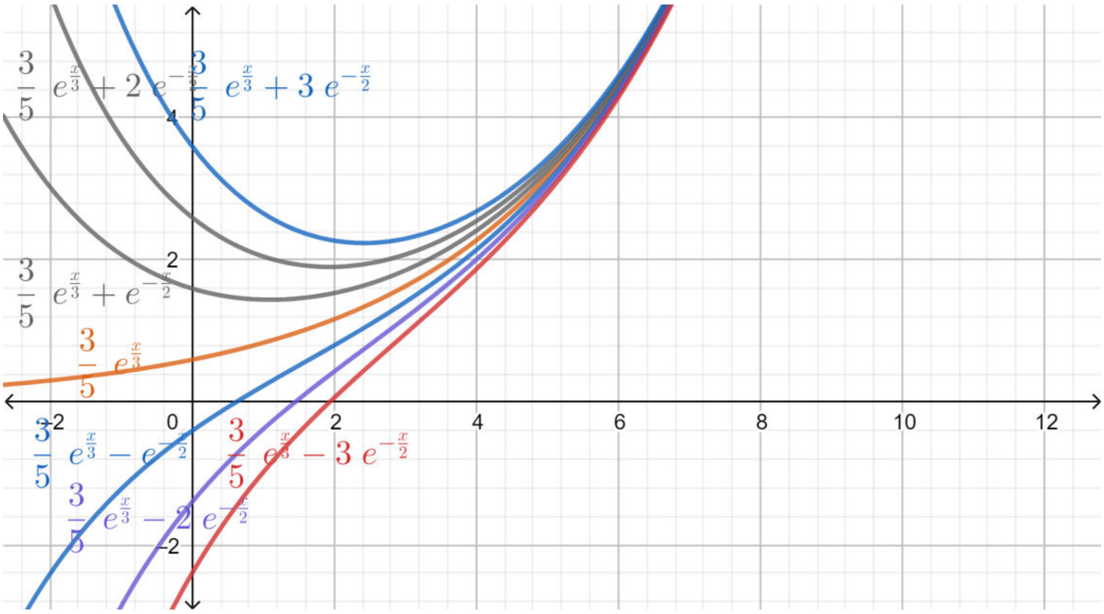
Therefore, the particular solution is

$$y = \frac{3}{5}e^{\frac{t}{3}} + \frac{2}{5}e^{-\frac{t}{2}}$$

As $t \to \infty, y(t) \to \infty$. And as $t \to \infty$, the term $\frac{2}{5}e^{-\frac{t}{2}} \to \infty$ and hence the integral curves are asymptotic to the curve

$$y = \frac{3}{5}e^{-\frac{t}{3}}.$$

Graphs of Solution Curves



Separable Differential Equations and Their Applications

The most general first order differential equation in normal form is:

$$\frac{dy}{dx} = f(x, y) \quad \dots \quad (1)$$

If (1) is non-linear there is no universal method to solve it.

Equation (1) can be put in the standard form

$$N(x, y)\frac{dy}{dx} + M(x, y) = 0 \quad \dots \quad (2)$$

where, $M(x, y) = -f(x, y)$ and $N(x, y) = 1$.

If $M(x, y)$ is a function of x alone and $N(x, y)$ is a function of y alone, then (2) reduces to

$$\frac{dy}{dx}N(y) + M(x) = 0$$

$$or, \qquad M(x)dx + N(y)dy = 0 \quad \dots \quad (3)$$

A differential equation is said to be separable or to have separable variables if the differential equation can be put in the form (3) i.e.

$$Mdx + Ndy = 0.$$

where, M is a function of x alone and N is a function of y alone.

Example 1 Show that the differential equation $\frac{dy}{dx} = \frac{x^2}{1-y^2}$ is seperable and find an equation for it's integral curves.

Solution

The given differential equation is

$$\frac{dy}{dx} = \frac{x^2}{1-y^2} \quad \dots \quad (1)$$

It can be written as

$$(1-y^2)dy = x^2dx$$
$$\Rightarrow x^2dx - (1-y^2)dy = 0$$

which, shows that (1) is seperable where $M(x) = x^2$ and $N(y) = 1 - y^2$.

Integrating it, we get

$$\int x^2\,dx - \int (1-y^2)\,dy = 0$$
$$or, \quad \frac{x^3}{3} - y + \frac{y^3}{3} = c_1$$
$$or, \quad x^3 - 3y + y^3 = c$$
$$or, \quad x^3 + y^3 - 3y = c$$

where, C is an arbitrary constant.

Example 2. Solve $(1+x)dy - ydx = 0$

Solution

The given differential equation is

$$(1+x)dy - ydx = 0 \quad \dots \quad (1)$$

It can be written as

$$\frac{dy}{y} - \frac{dx}{1+x} = 0.$$

Integrating, we get

$$\int \frac{dy}{y} - \int \frac{dx}{1+x} = 0$$
$$\Rightarrow \ln|y| - \ln|1+x| = lnc_1$$
$$\Rightarrow \ln\left|\frac{y}{1+x}\right| = lnc_1$$
$$\Rightarrow \left|\frac{y}{1+x}\right| = c_1$$
$$\Rightarrow \frac{y}{1+x} = \pm c_1$$
$$\Rightarrow y = c(1+x)$$

Therefore, the the general solution of (1) is

$$y = c(1+x)$$

Otherwise,

(1) can also be written as

$$(1+x)dy = ydx$$
$$\Rightarrow \frac{dy}{y} = \frac{dx}{1+x}$$
$$\Rightarrow \ln|y| = \ln|1+x| + c_1$$

$$\Rightarrow \mathrm{y} = e^{\ln|1+x|+c_1}$$
$$\Rightarrow \mathrm{y} = e^{\ln|1+x|}.e^{c_1}$$
$$\Rightarrow \mathrm{y} = (1+x).\mathrm{c}$$

Therefore, the general solution of (1) is

$$\mathrm{y} = c(1+x).$$

Example 3. Solve the IVP $\frac{dy}{dx} = -6xy, y(0) = 7$

Solution

The given IVP is

$$\frac{dy}{dx} = -6xy, y(0) = 7 \qquad \dots \qquad (1)$$

First, we find the general solution of

$$\frac{dy}{dx} = -6xy$$

Now,

$$\frac{dy}{dx} = -6xy$$
$$\Rightarrow \frac{dy}{y} = -6xdx$$

Integration implies

$$\int \frac{dy}{y} = -6\int xdx$$
$$\ln|y| = -3x^2 + c$$
$$\Rightarrow |y| = e^{-3x^2+c} = e^{-3x^2}.e^c$$

Using $y(0) = 7$, we get

$$7 = e^0.e^c$$
$$\Rightarrow e^c = 7 \Rightarrow c = ln7$$

Therefore, the required solution of the IVP (1) is

$$y = e^{-3x^2}.e^{ln7} = 7e^{-3x^2}.$$

Example 4. Solve the initial value problem

$$y' = \frac{3x^2 - e^x}{2y-5}, y(0) = 1$$

Solution

The given IVP is

$$\frac{dy}{dx} = \frac{3x^2 - e^x}{2y-5}, y(0) = 1 \qquad \dots \qquad (1)$$

Now,

$$\frac{dy}{dx} = \frac{3x^2 - e^x}{2y-5}$$
$$\Rightarrow (2y-5)dy = (3x^2 - e^x)dx$$

Integration implies

$$\int (2y-5)dy = \int (3x^2 - e^x)dx$$
$$\Rightarrow y^2 - 5y = x^3 - e^x + c$$

Therefore, the general solution is

$$y^2 - 5y = x^3 - e^x + c \qquad \dots \qquad (2)$$

Using $y(0) = 1$, we get

$$1 - 5 = 0 - 1 + c$$
$$\Rightarrow c = -3$$
$$\therefore\ c = -3$$

Hence, the solution of IVP (1) is

$$y^2 - 5y = x^3 - e^x - 3$$

To write it in explicit form

$$y^2 - 5y - x^3 + e^x + 3 = 0$$

which is quadratic in y.

$$y = \frac{5 \pm \sqrt{25 - 4.1.(-x^3 + e^x + 3)}}{2}$$
$$y = \frac{5 \pm \sqrt{25 + 4x^3 - 4e^x - 12}}{2}$$
$$y = \frac{5 \pm \sqrt{13 + 4x^3 - 4e^x}}{2}$$
$$\therefore\ y = \frac{5}{2} \pm \sqrt{x^3 - e^x + \frac{13}{4}} \qquad \ldots \qquad (3)$$

As $x = 0, y = 1$, we have

$$1 = \frac{5}{2} \pm \sqrt{0 - 1 + \frac{13}{4}}$$
$$\Rightarrow 1 = \frac{5}{2} \pm \frac{3}{2}$$

which is true for $-ve$ sign.

Hence, the required solution of the given IVP is

$$y = \frac{5}{2} - \sqrt{x^3 - e^x + \frac{13}{4}}$$

Example 5. Solve the initial value problem

$$y' = 2(1 + x)(1 + y^2);\ y(0) = 0$$

and determine where the solution attains its minimum value.

Solution

The given IVP is

$$y' = 2(1 + x)(1 + y^2);\ y(0) = 0 \qquad \ldots \qquad (1)$$

First, we find the general solution of

$$\frac{dy}{dx} = 2(1 + x)(1 + y^2)$$
$$\frac{dy}{(1 + y^2)} = 2(1 + x)dx$$

Integration implies

$$\int \frac{dy}{(1 + y^2)} = 2\int (1 + x)dx$$
$$\Rightarrow tan^{-1}y = 2x + x^2 + c$$

Using the initial condition $y(0) = 0$, we have $0 = 0 + c$

$$\therefore c = 0$$

Therefore, the solution of the given IVP is

$$tan^{-1}y = 2x + x^2$$
$$\Rightarrow y = tan(2x + x^2)$$

For minimum value of the solution

$$\frac{dy}{dx} = 0$$
$$\Rightarrow x = -1$$

Hence, the solution attains minimum value at $x = -1$.

Example 6. Solve the IVP $y' = \frac{2cos2x}{3+2y}, y(0) = -1$ and determine where the solution attains it's maximum value.

General solution

$$3y + y^2 = sin2x + c$$
$$y(0) = -1 \;\Rightarrow\; c = -2$$

and explicit solution is $y = -\frac{3}{2} - \sqrt{sin2x + \frac{1}{4}}$.

$\frac{dy}{dx} = 0 \;\Rightarrow x = \frac{\pi}{4}$, at which the solution has maximum value.

Example 7. Solve the IVP $y' = xy^3(1 + x^2)^{-\frac{1}{2}}$, $y(0) = 1$ and find the solution in explicit form. Also, determine the interval in which the solution is defined.

Solution

The given IVP is

$$y' = xy^3(1 + x^2)^{-\frac{1}{2}}, \qquad y(0) = 1 \qquad \ldots \qquad (1)$$

General solution is

$$-\frac{y^{-2}}{2} = \sqrt{1 + x^2} + c$$

Explicit solution is

$$y = \left[3 - 2\sqrt{1 + x^2}\right]^{-\frac{1}{2}}, using\ y(0) = 1.$$

The solution is defined for

$$3 - 2\sqrt{1 + x^2} \geq 0$$
$$\Rightarrow 3 \geq 2\sqrt{1 + x^2}$$
$$\Rightarrow 2\sqrt{1 + x^2} \leq 3$$
$$\Rightarrow 4(1 + x^2) \leq 9$$
$$\Rightarrow 4x^2 \leq 5 \Rightarrow x^2 \leq \frac{5}{4}$$
$$\Rightarrow |x| \leq \frac{\sqrt{5}}{2} \Rightarrow x \leq \pm\frac{\sqrt{5}}{2}$$

Modeling With First Order Differential Equations

Example 1. A tank initially contains 120 litres of pure water. A mixture contening a concentration of β g/L salt entres the tank at a rate of 2L/min, and the well stirred mixture leaves the tank at the same rate. Find an expression in terms of β for the amount of salt in the tank at any time t. Also, find the limiting amount of salt in the tank as $t \to \infty$.

Solution

Let $Q(t)$ be the amount of salt (in lb) in the tank in time t (min.)

The rate of change of salt in the tank $\frac{dQ}{dt}$ is given by

$$\frac{dQ}{dt} = \, flow\ in - flow\ out.$$

We know,

$$Flow\ in = Concentration \times rate\ of\ flow\ in$$

$$\frac{\beta gal}{1L} \times \frac{2L}{1min} = 2\beta gal/min.$$

And

$$flow\ out = concentration\ out \times rate\ of\ flow\ out$$

$$= \frac{Q(t)gal}{V(t)L} \times \frac{2L}{1min} = \frac{2Q(t)gal}{V(t)}/min.$$

Therefore,

$$\frac{dQ}{dt} = \, 2\beta - \frac{2Q}{V(t)} \qquad \ldots \qquad (1)$$

Here, $V(t)$ is the volume of mixture in the tank in time t. Then,

$$\frac{dV}{dt} = 2L/min - 2L/min = 0$$

$$\Longrightarrow dV = 0$$

Integration implies

$$V = c$$

Using initial condition, at $t = 0, V = 120$, we get $c = 120$.

$$\therefore V = 120$$

Hence, (1) reduces to

$$\frac{dQ}{dt} = \, 2\beta - \frac{2Q}{120} = 2\beta - \frac{Q}{60}$$

$$\frac{dQ}{2\beta - \frac{Q}{60}} = \, dt$$

Integraton implies

$$\int \frac{dQ}{(2\beta - \frac{Q}{60})} = \int dt$$

$$\Longrightarrow -60ln\left|2\beta - \frac{Q}{60}\right| = t - 60lnc$$

$$\Longrightarrow -60ln\left|2\beta - \frac{Q}{60}\right| + 60lnc = t$$

$$\Longrightarrow ln\left|2\beta - \frac{Q}{60}\right| - lnc = -\frac{t}{60}$$

$$\Longrightarrow ln\left|\frac{2\beta - \frac{Q}{60}}{c}\right| = -\frac{t}{60}$$

$$\Longrightarrow \left|\frac{2\beta - \frac{Q}{60}}{c}\right| = e^{-\frac{t}{60}}$$

$$\Longrightarrow \frac{2\beta - \frac{Q}{60}}{c} = e^{-\frac{t}{60}}, since\ e^{-\frac{t}{60}} > 0\ for\ all\ t.$$

$$\Rightarrow 2\beta - \frac{Q}{60} = ce^{-\frac{t}{60}}$$
$$\Rightarrow \frac{Q}{60} = 2\beta - ce^{-\frac{t}{60}}$$
$$\Rightarrow Q = 60\left[2\beta - ce^{-\frac{t}{60}}\right] \quad \ldots \quad (2)$$

Using initial contiion, where $t = 0, Q = 0$ (pure water), we have

$$0 = 60[2\beta - ce^{0}]$$
$$\Rightarrow 0 = [2\beta - c] \Rightarrow c = 2\beta.$$

Therefore, (2) reduces to

$$Q = 60\left[2\beta - 2\beta e^{-\frac{t}{60}}\right]$$
$$\Rightarrow Q = 120\beta\left[1 - e^{-\frac{t}{60}}\right]$$

In the limit, as $t \to \infty, 120\beta$.

Example 2. A 100-gallon tank with 50 gallons of pure water sits in a garage. A solution of 0.4 lbs salt per gallon enters the tank at a rate of 3 gallon per minute. The concentration leaves the tank at a rate of 2 gallon per minute. Find the selt concentration in the tank at any time t.

Solution

Let $Q(t)$ be the amount of salt in the tank at any time t. The rate of change of salt Q in the tank $\frac{dQ}{dt}$ as then given by

$$\frac{dQ}{dt} = flow\ in - flow\ out.$$

where,

$$flow\ in = (concentration\ in) \times rate\ of\ flow\ in$$
$$= \frac{0.4lb}{1gal} \times \frac{3gal}{1min} = 1.2lb/min.$$

And

$$flow\ out = (concentration\ out) \times rate\ of\ flow\ out$$
$$= \frac{Q(t)}{V(t)gal} \times \frac{2gal}{1min} = \frac{2Q(t)}{V(t)} lb/min.$$
$$\therefore \frac{dQ}{dt} = 1.2lb/min - \frac{2Q(t)}{V(t)} lb/\min \quad \ldots \quad (1)$$

If $V(t)$ is the volume of salt in the tank in time t, then

$$\frac{dV}{dt} = 3gal/min - 2gal/min = 1gal/min$$
$$\therefore \frac{dV}{dt} = 1$$
$$\Rightarrow dV = dt$$

Integration implies

$$V(t) = t + c.$$

Using initial condition as $t = 0, V = 50$, we get $c = 50$

$$\therefore V(t) = (t + 50)gal.$$

Now, equation (1) becomes

$$\frac{dQ}{dt} = 1.2 - \frac{2Q}{t + 50}$$

$$\frac{dQ}{dt} + \frac{2Q}{t+50} = 1.2 \quad \ldots \quad (2)$$

Equation (2) is linear with $P = \frac{2Q}{t+50}$.

$$\therefore\ I.F.\, e^{\int \frac{2}{t+50}dt} = e^{2\ln(t+50)} = e^{\ln(t+50)^2} = (t+50)^2$$

Multiplying both sides of (2) by $(t+50)^2$, we get

$$(t+50)^2\frac{dQ}{dt} + \ 2(t+50)Q = 1.2(t+50)^2$$

$$\Rightarrow \frac{d}{dt}[Q.(t+50)^2] = 1.2(t+50)^2$$

Integration implies

$$Q.(t+50)^2 = 1.2\int (t+50)^2\, dt$$

$$\Rightarrow Q(t+50)^2 = 1.2 \times \frac{(t+50)^3}{3} + c$$

Again, initially at $t = 0, Q = 0$ (being pure water), we have

$$0.(0+50)^2 = 1.2 \times \frac{(0+50)^3}{3} + c$$

$$\Rightarrow 0 = 0.4 \times 125000 + c \Rightarrow 0 = 50000 + c$$

$$\Rightarrow c = -50000$$

Therefore, the amount of salt at any time is

$$Q(t+50)^2 = 1.2 \times \frac{(t+50)^3}{3} - 50000$$

$$Q(t) = (t+50)^2 = \frac{2}{5}(t+50) - \frac{50000}{(t+50)^2}$$

Difference Between Linear and Non-linear Differential Equations:

Theorem 1 (Statements Only)

If the functions p and g are continuous on an open interval $I: \alpha < t < \beta$ containing the point $t = t_0$, then there exists a unique function $y = \phi(t)$ that satisfies the differential equation

$$y' + p(t)y = g(t) \quad \ldots \quad (1)$$

for each t in I and that also satisfies the initial condition.

$$y(t_0) = y_0 \quad \ldots \quad (2)$$

where, y_0 is an arbitrary prescribed initial value.

This theorem is called uniqueness theorem

Theorem 2 (Statement Only)

Let the functions f and $\frac{\partial f}{\partial y}$ be continuous in some rectangle $R: \alpha < t < \beta, \gamma < y < \delta$ containing the point (t_0, y_0). Then, in some interval $t_0 - h < t < t_0 + h$ contained in $\alpha < t < \beta$, there is a unique solution $y = \phi(t)$ of the initial value problem

$$y' = f(t, y), y(t_0) = y_0 \quad \ldots \quad (1).$$

Example 1. Determine the interval in which the solution of the given IVP is certain to exist

$$ty' + 2y = 4t^2\,; y(1) = 2.$$

Also, find the solution in that inval.

Solution

The given IVP is

$$ty' + 2y = 4t^2\,; y(1) = 2 \quad \ldots \quad (1)$$

It can also be written as

$$y' + \frac{2}{t}y = 4t \quad ... \quad (2)$$

Comparing it with

$$y' + p(t)y = g(t),$$

we have, $p(t) = \frac{2}{t}$ and $g(t) = 4t$.

Here, $g(t) = 4t$ is continuous for every value of $t \in R$. But, $p(t) = \frac{2}{t}$ is continuous for all $t \in R$ other than 0 i.e. $p(t)$ is discontinuous at $t = 0$.

Therefore, the solution (2) is certain to exist in the internal $(-\infty, 0) \cup (0, \infty)$ $i.e. t > 0$ and $t < 0$.

But, the initial value is given by $t = 1$ which belongs to the internal $(0, \infty)$. Hence, the IVP has unique solution in the internal $(0, \infty)$.

Next, to find the solution of (1).

We see that (2) is linear with

$$p(t) = \frac{2}{t} \text{ and } g(t) = 4t.$$

$$I.F. = e^{\int \frac{2}{t} dt} = e^{2lnt} = e^{lnt^2} = t^2.$$

$$\therefore \; I.F. = t^2$$

Multiplying (2) by t^2, we get

$$t^2 \frac{dy}{dt} + 2ty = 4t^3$$

$$\Rightarrow \frac{d}{dt}(t^2 y) = 4t^3$$

Integration implies

$$t^2 y = \int 4t^3 \, dt$$

$$t^2 y = t^4 + c \quad ... \quad (3)$$

Using $y(1) = 2$, we get

$$1^2 . 2 = 1^4 + c$$

$$\Rightarrow c = 1$$

Putting $c = 1$, (3) reduces to

$$t^2 y = t^4 + 1$$

$$\Rightarrow y = t^2 + \frac{1}{t^2}, t > 0$$

$\therefore$ The required solution of the IVP is

$$y = t^2 + \frac{1}{t^2} \; in \; (0, \infty).$$

Example 2. Solve the IVP $\frac{dy}{dx} = \frac{3x^2+4x+2}{2(y-1)}$; $y(0) = 1$ and determine the interval in which solution exists.

Solution

The given IVP is

$$\frac{dy}{dx} = \frac{3x^2 + 4x + 2}{2(y-1)} \quad ... \quad (1)$$

$$y(0) = 1 \quad ... \quad (2)$$

Comparing (1) with $\frac{dy}{dx} = f(x, y)$, we get

$$f(x, y) = \frac{3x^2 + 4x + 2}{2(y-1)}$$

Hence, f(x, y) is continuous for all $x \in R$ and all $y \neq 1$. That is, $f(x, y)$ is continuous for all $y \neq 1$.
Next,

$$\frac{\partial f}{\partial y} = \frac{\partial}{\partial y}\left[\frac{3x^2 + 4x + 2}{2(y-1)}\right] = -\frac{1}{2}\left[\frac{3x^2 + 4x + 2}{(y-1)^2}\right].$$

Which is continuous for $y \neq 1$.
Therefore, both $f(x, y)$ and $\frac{\partial f}{\partial y}$ are both continuous for $y \neq 1$.
Next, we solve (1)

$$\frac{dy}{dx} = \frac{3x^2 + 4x + 2}{2(y-1)}$$
$$\Rightarrow 2(y-1)dy = (3x^2 + 4x + 2)dx$$

Integration implies

$$2\int (y-1)dy = \int (3x^2 + 4x + 2)dx$$
$$\Rightarrow y^2 - 2y = x^3 + 2x^2 + 2x + c \qquad \ldots \qquad (3)$$

Using initial condition, $y(0) = 1$ we get

$$1^2 - 2.1 = 0 + 2.0 + 2.0 + c$$
$$\Rightarrow -1 = c$$

$\therefore c = -1$
Hence, (3) reduces to

$$y^2 - 2y = x^3 + 2x^2 + 2x - 1$$
$$\Rightarrow y^2 - 2y - x^3 - 2x^2 - 2x + 1 = 0$$

Therefore,

$$y = \frac{2 \pm \sqrt{4 + 4(x^3 + 2x^2 + 2x - 1)}}{2.1}$$
$$= 1 \pm \frac{\sqrt{4 + 4(x^3 + 2x^2 + 2x - 1)}}{2}$$
$$= 1 \pm \sqrt{x^3 + 2x^2 + 2x}$$

which exists for

$$x^3 + 2x^2 + 2x > 0$$
$$\Rightarrow x[x^2 + 2x + 2] > 0$$
$$\Rightarrow x[(x+1)^2 + 1] > 0$$
$$\Rightarrow x > 0.$$

Example 3. State where in the ty-plane the hypothesis of the theorem 2 are satisfied in the problem

$$\frac{dy}{dt} = \frac{1 + t^2}{3y - y^2}$$

Solution
The given differential equation is

$$\frac{dy}{dt} = \frac{1 + t^2}{3y - y^2} = \frac{1 + t^2}{y(3-y)} \qquad \ldots \qquad (1)$$

Also,

$$\frac{\partial f}{\partial y} = \frac{\partial}{\partial y}\left[\frac{1 + t^2}{y(3-y)}\right] = \frac{1 + t^2}{y(3-y)^2} - \frac{1 + t^2}{y^2(3-y)}$$

Here both (1) and (2) are discontinuous on ty-plane at $y = 0$ and $y = 3$. Hence, the required Region is $y \neq 0, y \neq 3$. Also, it's solution is

$$\frac{dy}{dt} = \frac{1+t^2}{3y-y^2} \Rightarrow (3y - y^2)dy = (1+t^2)dt$$

$$\Rightarrow \int (3y - y^2)\, dy = \int (1+t^2)\, dt$$

$$\Rightarrow \frac{3}{2}y^2 - \frac{y^3}{3} = t + \frac{t^3}{3} + c.$$

Bernouli's Equation

A Differential equation of the form

$$\frac{dy}{dt} + p(t)y = g(t)y^n \quad \ldots \quad (1)$$

is called Bernouli's equation.

To solve such an equation, we divide both sides of (1) by y^n to get

$$\frac{1}{y^n}\frac{dy}{dt} + \frac{p(t)}{y^{n-1}} = g(t) \quad \ldots \quad (2)$$

We put $z = \frac{1}{y^{n-1}} = y^{1-n}$

$$\Rightarrow \frac{dz}{dt} = (1-n)y^{-n}\frac{dy}{dt}$$

$$\Rightarrow \frac{1}{(1-n)}\frac{dz}{dt} = \frac{1}{y^n}\frac{dy}{dt}$$

which when substituted in (2), we get

$$\frac{1}{(1-n)}\frac{dz}{dt} + p(t)z = g(t).$$

$$\Rightarrow \frac{dz}{dt} + (1-n)p(t)z = (1-n)g(t) \quad \ldots \quad (3)$$

which is now linear in z. We now follow the same procedure as before to solve $\frac{dy}{dt} + p(t)y = g(t)$.

Example 1 Solve $\frac{dy}{dx} + xsin2y = x^3cos^2y$.

Solution

The given equation is

$$\frac{dy}{dx} + xsin2y = x^3cos^2y \quad \ldots \quad (1)$$

which is Bernouli's equation.

Now, (1) can be written as

$$sec^2y\frac{dy}{dx} + x\frac{sin2y}{cos^2y} = x^3$$

$$or, \qquad sec^2y\frac{dy}{dx} + x\frac{2sinycosy}{cos^2y} = x^3$$

$$or, \qquad sec^2y\frac{dy}{dx} + 2xtany = x^3 \quad \ldots \quad (2)$$

Put $tany = t$.

$$\therefore\ sec^2y\frac{dy}{dx} = \frac{dt}{dx}$$

Thus, (2) reduces to

$$\frac{dt}{dx} + 2xt = x^3 \quad \ldots \quad (3)$$

which is linear with $p(x) = 2x, g(x) = x^3$.

$\therefore I.F. = e^{\int p dx} = e^{\int 2x dx} = e^{x^2}$.

Multiplying (3) by e^{x^2}, we get

$$e^{x^2}\frac{dt}{dx} + 2xe^{x^2}t = x^3e^{x^2}$$

$$\frac{d}{dx}\left(te^{x^2}\right) = x^3e^{x^2}$$

Integration implies

$$te^{x^2} = \int x^3e^{x^2}dx = \int x\,.x^2e^{x^2}dx$$

Put $z = x^2 \Longrightarrow dz = 2xdx \Longrightarrow \frac{1}{2}dz = 2xdx$.

Therefore,

$$RHS = \int ze^z\frac{1}{2}dz = \frac{1}{2}\int ze^zdz$$

$$= \frac{1}{2}[ze^z - e^z] + c$$

$$= \frac{1}{2}\left[x^2e^{x^2} - e^{x^2}\right] + c$$

Hence,

$$te^{x^2} = \frac{1}{2}\left[x^2e^{x^2} - e^{x^2}\right] + c$$

$$\Longrightarrow t = \frac{1}{2}[x^2 - 1] + ce^{-x^2}$$

$$\Longrightarrow tany = \frac{1}{2}[x^2 - 1] + ce^{-x^2}$$

Autonomous Equations and Population Dynamics (Stability Theory)

Autonomous Differential equation

An ordinary differential equation in which $\frac{dy}{dx}$ is a function of y alone is called an Autonomous differential equation.

An autonomous first order differential equation is of the form

$$f(y, y') = 0 \ or \ \frac{dy}{dx} = f(y).$$

For example,

i. $\frac{dy}{dx} = y(3 - y)$ is Autonomous, where $f(y) = y(3 - y)$

where as

ii. $\frac{dy}{dx} = y^2 + x^2$ is not Autonomous wehre, $y^2 + x^2 = f(x, y)$.

Solving Autonomous Equation

An autonomous differential equation is seperable. That is, $\frac{dy}{dx} = f(y)$ can be written as

$$\frac{dy}{f(y)} = dx \Longrightarrow \int\frac{dy}{f(y)} = x + c$$

Equilibria of Autonomous Equations

The equilibria or the constant solutions of the autonomous defferential equations $\frac{dy}{dx} = f(y)$ are the roots of the equation $\frac{dy}{dx} = 0$, that is, of the equation $f(y) = 0$.

A real constant c is called critical point/equilibrium point/rest point of $\frac{dy}{dx} = f(y)$ if $f(c) = 0$. Thus, if c is a critical point of $\frac{dy}{dx} = f(y)$, then $y = c$ is a constant solution/equilibrium solution of the autonomous differential equation $\frac{dy}{dx} = f(y)$.

Phase Line

A line drawn parallel to y −axis which shows equilibrium values, the intervals and the behavior of an autonomons differential equation $\frac{dy}{dx} = f(y)$, is called the phase line.

Sample Phase Line

Example $\frac{dy}{dx} = (1 - y)(3 - y)$

Classification of Equilibrium Solutions (Stable, Unsatable and Semistable)

Stable Solution

It all integral curves coverage towards an equilibrium solution as x increases, then the equilibrium solution is said to be stable or asymptotically stable. That's, an equilibrium solution $y = c$ is satable if all solutions with initial condition y_0 close to c approach c as $x \to \infty$.

Unstable Solution

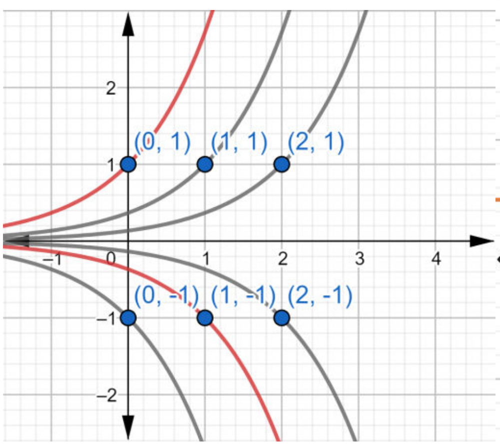
If all integral curves diverge away from an equilibrium solution as $x \to \infty$, then the equilibrium solution $y = c$ is said to be unstable.

Semistable Solution

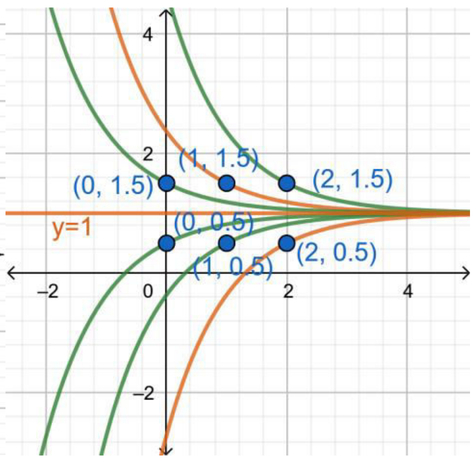
An equilibrium solution $y = c$ is said to be semi-stable if the solution curves on one side of $y = c$ converage to c as $x \to \infty$ and the integral curves on the other side of $y = c$ diverage away from c.

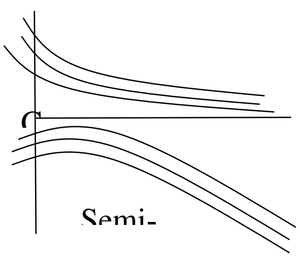
GRAPH

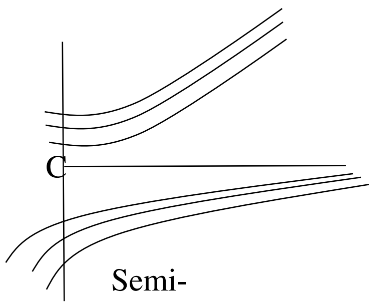
Unstable



Stable







Semi-

Example 1. Find and classify the equilibirium points of

$$y' = (y+1)(y-2).$$

Solution

The given equation is

$$y' = (y+1)(y-2) \quad ... \quad (1)$$

For equilibrium points $\frac{dy}{dt} = 0$

$$\Rightarrow (y+1)(y-2) = 0$$

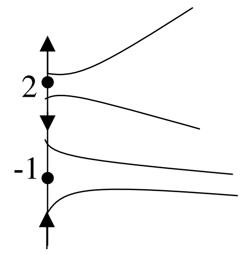
$$\therefore\ y = -1, 2$$

Therefore, the equilibrium solution are

$$y = -1 \ and \ y = 2$$

These points divide the real line into three intervals $(-\infty, -1), (-1,2), (2, \infty)$.

Intervals	$y+1$	$y-2$	$(y+1)(y-2)$
$(-\infty, -1)$	< 0	< 0	> 0
$(-1,2)$	> 0	< 0	< 0
$(2, \infty)$	> 0	> 0	> 0



Therefore, $\frac{dy}{dt} > 0$ for $y < -1$, that is, $y(t)$ increasing for $y < -1$.

And, $\frac{dy}{dt} < 0$ for $-1 > y < 2$, that is, $y(t)$ decreasing for $-1 > y < 2$.

Also, $\frac{dy}{dt} > 0$ for $y > 2$, that is, $y(t)$ increasing for $y > 2$.

Hence, the equilibrium solution $y = -1$ is stable and $y = 2$ is unstable.

Example 2. Sketch the graph of $f(y)$ versus y, determine the critical points and classify each of them as asymptotically stable or unstable or semi-stable. Draw phase line and graphs of several integral curves in the ty −plane for $y' = y(4-y^2), -\infty < y < \infty$.

Solution

The given autonomous differential equation is

$$y' = y(4-y^2) = y(2-y)(2+y) \quad ... \quad (1)$$

where $f(y) = y(2-y)(2+y) \quad ... \quad (2)$

$$\therefore\ \frac{d^2y}{dt^2} = \frac{d}{dt}\left(\frac{dy}{dt}\right) = \frac{df(y)}{dt} = \frac{df(y)}{dy} \times \frac{dy}{dt} = f'(y)f(y) \quad ... \quad (3)$$

And

$$f'(y) = 4 - 3y^2 \quad ... \quad (4)$$

For equilibrium points, we have

$$\frac{dy}{dt} = 0 \Rightarrow y(2-y)(2+y) = 0$$

$\therefore y = 0, 2, -2$

Therefore, the equilibrium solutions are

$$y = 0, y = 2 \ and \ y = -2.$$

For points of inflection, $f'(y) = 0$

$$\Rightarrow 4 - 3y^2 = 0$$

$\therefore y = \frac{2}{\sqrt{3}}, -\frac{2}{\sqrt{3}}.$

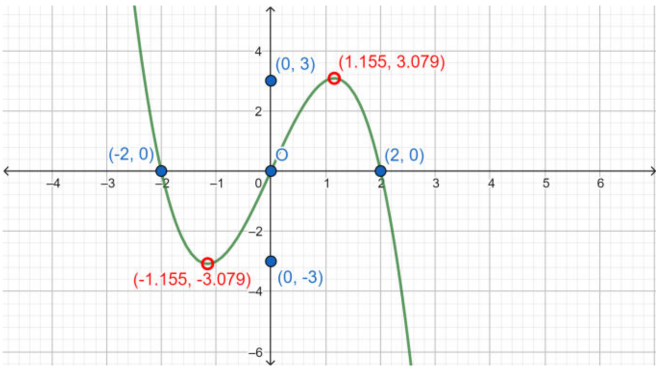
Now we observe the behaviours of $f(y), f'(y)$ and $\frac{d^2y}{dt^2} = f'(y)f(y)$ in the intervals $(-\infty, -2), \left(-2, -\frac{2}{\sqrt{3}}\right), \left(-\frac{2}{\sqrt{3}}, 0\right), \left(0, \frac{2}{\sqrt{3}}\right), \left(\frac{2}{\sqrt{3}}, 2\right) and\ (2, \infty)$.

Table

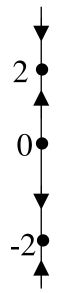
Intervals	$f(y)$	$f'(y)$	$\frac{d^2y}{dt^2} = f'(y)f(y)$	Bahaviour of $y(t)$
$(-\infty, -2)$	>0	<0	<0	increasing/concave down
$\left(-2, -\frac{2}{\sqrt{3}}\right)$	<0	<0	>0	Decreasing/concave up
$\left(-\frac{2}{\sqrt{3}}, 0\right)$	<0	>0	<0	Decreasing/concave down
$\left(0, \frac{2}{\sqrt{3}}\right)$	>0	>0	>0	Decreasing/concave up
$\left(\frac{2}{\sqrt{3}}, 2\right)$	>0	<0	<0	Decreasing/concave down
$(2, \infty)$	<0	<0	>0	Decreasing/concave up

From the above table we see that the integral curvep $y(t)$ near $y = 0$ are increasing above and decreasing below. So, the solution $y = 0$ is unstable. Near $y = -2$, the integral curves are increasing below and decreasing above. So, the solution $y = -2$ is stable. And near $y = 2$, the solution curves are increasing below and decreasing above. Therefore, the solution $y = 2$ is stable. Moreover, using the information in the above table the phase line, the graph of $f(y)$ versus y and the graph of $y(t)$ versus t are drawn as follows:

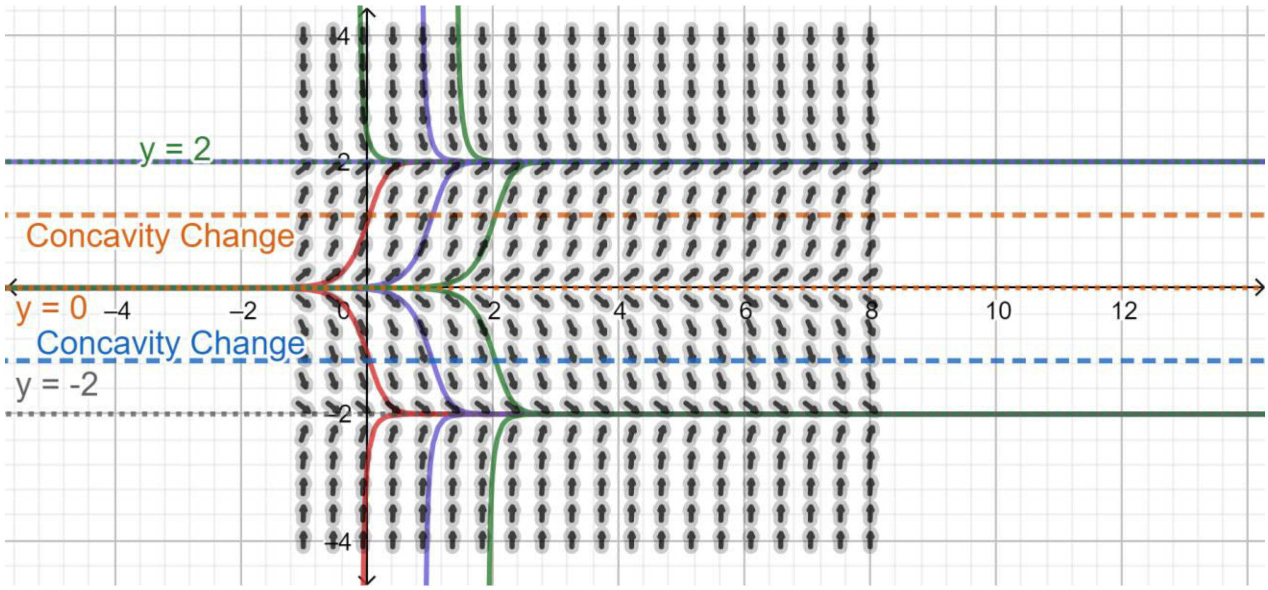
Figure



Graph of f(y) Versus y



Phase Line of $\frac{dy}{dt}$



Graph of y(t) Versus t

Example 3. Given the differential equation $\frac{dy}{dt} = y^2(1-y)^2, -\infty < y < \infty$, determine the critical points, classify the equilibrium solutions as stable, unstable or semi-stable, draw the phase line, sketch the graph of $f(y)$ versus y and sketch so graph of solution (integral) curves in ty −plane.

Solution

The given autonomous differential equation is

$$\frac{dy}{dt} = y^2(1-y)^2, -\infty < y < \infty \quad \ldots \quad (1)$$

where $f(y) = y^2(1-y)^2 \quad \ldots \quad (2)$

$$\therefore \frac{d^2y}{dt^2} = \frac{d}{dt}\left(\frac{dy}{dt}\right) = \frac{df(y)}{dt} = \frac{df(y)}{dy} \times \frac{dy}{dt} = f'(y)f(y) \quad \ldots \quad (3)$$

And

$$f'(y) = 2y(1-3y+2y^2) = 2y(2y-1)(y-1) \quad \ldots \quad (4)$$

For equilibrium points, we have

$$\frac{dy}{dt} = 0 \Longrightarrow y^2(1-y)^2 = 0$$

$\therefore y = 0, 1.$

Therefore, the equilibrium solutions are

$$y = 0 \; and \; y = 1.$$

For points of inflection,

$$f'(y) = 0$$
$$\Longrightarrow 2y(2y-1)(y-1) = 0$$

$\therefore y = 0, \frac{1}{2}, 1.$

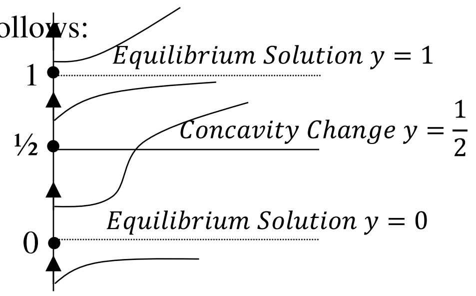
Now we observe the behaviours of $f(y), f'(y)$ and $\frac{d^2y}{dt^2} = f'(y)f(y)$ in the intervals $(-\infty, 0), \left(0, \frac{1}{2}\right), \left(\frac{1}{2}, 1\right) \; and \; (1, \infty)$.

Table

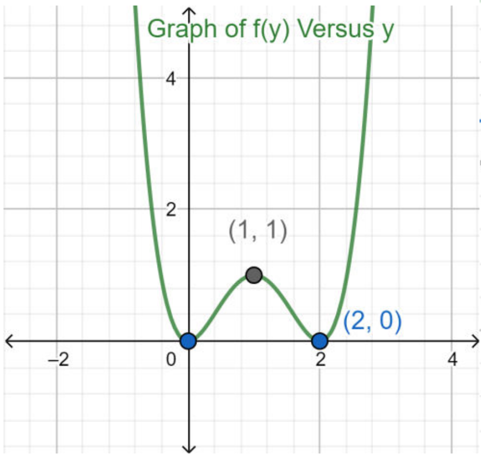
Intervals	$f(y)$	$f'(y)$	$\frac{d^2y}{dt^2} = f'(y)f(y)$	Bahaviour of $y(t)$
$(-\infty, 0)$	>0	<0	<0	Increasing/Concave Down
$\left(0, \frac{1}{2}\right)$	>0	>0	>0	Increasing/Concave Up
$\left(\frac{1}{2}, 1\right)$	>0	<0	<0	Increasing/Concave Down
$(1, \infty)$	>0	>0	>0	Increasing/Concave Up

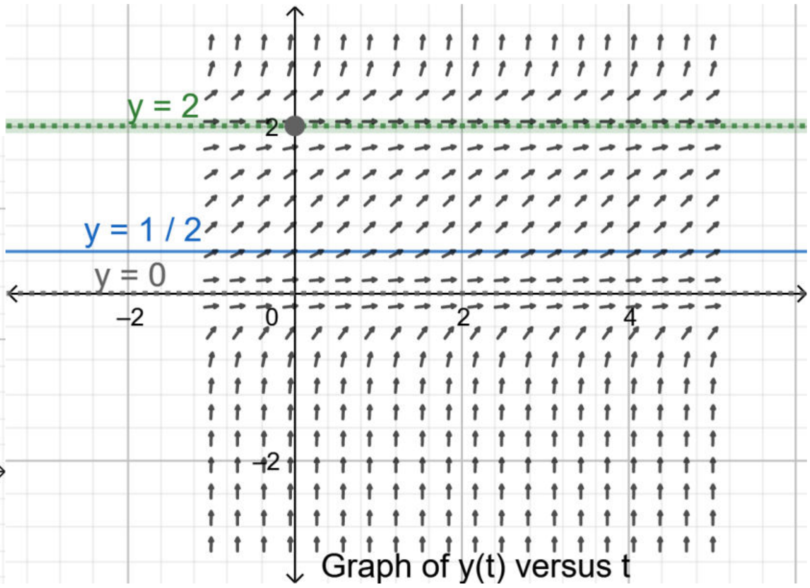
From the above table we see that the integral curvep $y(t)$ near $y = 0$ are increasing below and increasing above. So, the solution $y = 0$ is unstable. Near $y = \frac{1}{2}$, the integral curves are increasing above and increasing down. So, the solution $y = \frac{1}{2}$ is unstable. And near $y = 1$, the solution curves are increasing below and increasing above. Therefore, the solution $y = 1$ is semi-stable.

Moreover, using the information in the above table the phase line, the graph of $f(y)$ versus y and the graph of $y(t)$ versus t are drawn as follows:



Phase Line of $\frac{dy}{dt}$





Exact Differential Equation

A differential equation $Mdx + Ndy = 0$, where M and N are functions of x and y, is called exact if there exists a funetion $f(x, y)$ such that

$$d[f(x, y)] = Mdx + Ndy$$

That is,

$$\frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy = Mdx + Ndy.$$

Example 1. The differential equation $y^2dx + 2xydy = 0$ ... (1)

is an exact differential equation since there is a function $f(x, y) = xy^2$ such that

$$d(xy^2) = \frac{\partial}{\partial x}(xy^2)dx + \frac{\partial}{\partial y}(xy^2)dy$$

$$= y^2dx + 2xydy = Mdx + Ndy.$$

Therefore, $y^2dx + 2xydy = 0$ may be written as

$$d(xy^2) = 0$$

which when integrated gives

$$xy^2 = c$$

Hence, the general solution of (1) is

$$xy^2 = c.$$

Theorem The necessary and sufficient condition for the differential equation $Mdx + Ndy = 0$ to be exact is $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$.

Working Rule for Solving and Exact Differential Equation

Step 1 Integrate M with respect to x keeping y constant

Step 2 Integrate the terms of N free of x with respect to y.

Step 3 Equate the sum of the terms obtained in Steps (1) and (2) to an arbitrary constant c.

$$i.e. \int Mdx + \int (terms\ of\ N\ free\ from\ x)dy = c$$

Example 1 Solve $(x^2 - 4xy - 2y^2)dx + (y^2 - 4xy - 2x^2)dy = 0$

Solution

The given differential equation is

$$(x^2 - 4xy - 2y^2)dx + (y^2 - 4xy - 2x^2)dy = 0 \quad ... \quad (1)$$

Comparing it with

$$Mdx + Ndy = 0 \quad ... \quad (2)$$

we have

$$M = x^2 - 4xy - 2y^2 \ and \ N = y^2 - 4xy - 2x^2$$

$$\frac{\partial M}{\partial y} = -4x - 4y, \qquad \frac{\partial N}{\partial y} = -4y - 4x,$$

which imply $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial y}$.

Therefore, equation (1) is exact. So, the solution is

$$\int Mdx + \int (terms\ of\ N\ free\ of\ x)dy = c$$

$$\Rightarrow \int (x^2 - 4xy - 2y^2)dx + \int y^2 dy = c_1$$

$$\Rightarrow \frac{x^3}{3} - 2x^2y - 2xy^2 + \frac{y^3}{3} = c_1$$

$$\Rightarrow x^3 + y^3 - 6x^2y - 6xy^2 = c$$

Hence, the required solution is

$$x^3 + y^3 - 6xy(x + y) = c.$$

where c is an arbitrary constant.

Eamples

1. **Solve** $\left(1 + e^{\frac{x}{y}}\right) dx + e^{\frac{x}{y}}\left(1 - \frac{x}{y}\right) dy = 0$
2. **Solve** $\left(y^2 e^{xy^2} + 4x^3\right)dx + \left(2xye^{xy^2} - 3y^2\right)dy = 0$

Integrating Factor If a differential equation $Mdx + Ndy = 0$ is not exact, it can be made exact by multiplying by some function of x and y. Such a function is called an integrating factor $I.F.$

Rule I

Integoating Factor by Inspestion

$(i) \quad d\left(\frac{y}{x}\right) = \frac{xdy - ydx}{x^2} \qquad (ii) \quad d\left(\frac{x}{y}\right) = \frac{ydx - xdy}{y^2}$

$(iii) \quad d\left(\frac{y^2}{x}\right) = \frac{2xydy - y^2dx}{x^2} \qquad (iv) \quad d\left(\frac{x^2}{y}\right) = \frac{2yxdx - x^2dy}{y^2}$

$(v) \quad d\left(\frac{y^2}{x^2}\right) = \frac{2x^2ydy - 2xy^2dx}{x^4} \qquad (vi) \quad d\left(\frac{x^2}{y^2}\right) = \frac{2xy^2dx - 2yx^2dy}{y^4}$

$(vii) \quad d(\ln(xy)) = \frac{xdy + ydx}{xy} \qquad (viii) \quad d\left(\ln\left(\frac{y}{x}\right)\right) = \frac{xdy - ydx}{xy}$

$(ix) \quad d\left(\ln\left(\frac{x}{y}\right)\right) = \frac{ydx - xdy}{xy} \qquad (x) \quad d\left[\frac{1}{2}\ln(x^2 + y^2)\right] = \frac{xdx + ydy}{x^2 + y^2}$

$(xi) \quad d\left(-\frac{1}{xy}\right) = \frac{xdy + ydx}{x^2y^2} \qquad (xii) \quad d\left(\frac{e^x}{y}\right) = \frac{ye^xdx + e^xdy}{y^2}$

$(xiii) \quad d(xy) = xdy + ydx \qquad (xiv) \quad d\left(tan^{-1}\frac{y}{x}\right) = \frac{xdy - ydx}{x^2 + y^2}$

$(xv) \quad d\left(tan^{-1}\frac{x}{y}\right) = \frac{ydx - xdy}{x^2 + y^2} \qquad (xvi) \quad d(sin^{-1}xy) = \frac{xdy + ydx}{\sqrt{1 - x^2y^2}}$

Example 1. Solve $ydx - xdy + (1 + x^2)dx + x^2 sinydy = 0$

Solution

The given equation is

$$ydx - xdy + (1 + x^2)dx + x^2 sinydy = 0 \quad ... \quad (1)$$

Dividing both sides of (1) by x^2, we get

$$\frac{ydx - xdy}{x^2} + \left(1 + \frac{1}{x^2}\right) dx + sinydy = 0$$

$$or, \quad -\left[\frac{xdy - ydx}{x^2}\right] + \left(1 + \frac{1}{x^2}\right) dx + sinydy = 0$$

$$or, \quad -d\left(\frac{y}{x}\right) + \left(1 + \frac{1}{x^2}\right) dx + sinydy = 0$$

Integration implies

$$-\frac{y}{x} + \left(x - \frac{1}{x}\right) - cosy = c$$

$$or, \quad -y + x^2 - 1 - xcosy = cx$$

Therefore, the required solution is

$$-y + x^2 - 1 - xcosy = cx$$

where c is some orbitrary constant.

Example 2. Solve $(x^3 + xy^2 + a^2y)dx + (y^3 + yx^2 - a^2x)dy = 0$

Solution

The Given equation is

$$(x^3 + xy^2 + a^2y)dx + (y^3 + yx^2 - a^2x)dy = 0 \quad ... \quad (1)$$

$$\Rightarrow x(x^2 + y^2)dx + a^2ydx + y(y^2 + x^2)dy - a^2xdy = 0$$

$$\Rightarrow x(x^2 + y^2)dx + y(y^2 + x^2)dy + a^2(ydx - xdy) = 0$$

$$\Rightarrow xdx + ydy + a^2\left(\frac{ydx - xdy}{x^2 + y^2}\right) = 0$$

$$\Rightarrow d\left(\frac{x^2}{2}\right) + d\left(\frac{y^2}{2}\right) + a^2 d\left(tan^{-1}\frac{x}{y}\right) = 0$$

Integration implies

$$\frac{x^2}{2} + \frac{y^2}{2} + a^2 tan^{-1}\frac{x}{y} = c_1$$

$$x^2 + y^2 + 2a^2 tan^{-1}\frac{x}{y} = c$$

Hence, the required solution is

$$x^2 + y^2 + 2a^2 tan^{-1}\frac{x}{y} = c$$

where c is an arbitrary constant.

Rule II

If the equation $\boldsymbol{Mdx + Ndy = 0}$ is homogeneous and $\boldsymbol{Mx + Ny \neq 0}$, then $\boldsymbol{I.F. = \frac{1}{Mx+Ny}}$.

Example 1. Solve $(x^2y - 2xy^2)dx - (x^3 - 3x^2y)dy = 0$

Solution

The given equation is

$$(x^2y - 2xy^2)dx - (x^3 - 3x^2y)dy = 0 \quad ... \quad (1)$$

Clearly (1) is homogeneous differential equation, where

$$M = (x^2y - 2xy^2), N = -(x^3 - 3x^2y)$$

$$\therefore \; Mx + Ny = (x^2y - 2xy^2)x - (x^3 - 3x^2y)y$$

$$= x^3y - 2x^2y^2 - x^3y + 3x^2y^2 = x^2y^2 \neq 0$$

$$\therefore I.F. = \frac{1}{Mx+Ny} = \frac{1}{x^2y^2}$$

Multiplying (1) by $I.F. = \frac{1}{x^2y^2}$, we get

$$\left(\frac{1}{y} - \frac{2}{x}\right)dx - \left(\frac{x}{y^2} - \frac{3}{y}\right)dy = 0$$

which is now exact. So it's solution is

$$\int \left(\frac{1}{y} - \frac{2}{x}\right)dx + \int \frac{3}{y}dy = c_1$$

$$\Rightarrow \frac{x}{y} - 2lnx + 3lny = lnc$$

$$\Rightarrow \frac{x}{y} - lnx^2 + lny^3 = lnc$$

$$\Rightarrow -lnx^2 + lny^3 - lnc = -\frac{x}{y}$$

$$\Rightarrow ln\frac{y^3}{cx^2} = -\frac{x}{y}$$

$$\Rightarrow \frac{y^3}{cx^2} = e^{-\frac{x}{y}}$$

$$\Rightarrow y^3 = cx^2e^{-\frac{x}{y}}$$

Hence, the required solution is

$$y^3 = cx^2e^{-\frac{x}{y}}$$

where c is some arbitrary constant.

Example 2. Solve $x^2ydx - (x^3 - y^3)dy = 0$

Rule III

If the equation $Mdx + Ndy = 0$ is of the form

$$yf(xy)dx + xg(xy)dy = 0$$

then integrating factor is

$I.F. = \frac{1}{Mx-Ny}$ of $Mdx + Ndy = 0$, provided $Mx - Ny \neq 0$

Example 2. Solve $(xysinxy + cos\, xy)ydx + (xysinxy - cosxy)xdy = 0$

Solution

The given differential equation is

$$(xysinxy + cos\, xy)ydx + (xysinxy - cosxy)xdy = 0 \quad \ldots \quad (1)$$

which is of the form $Mdx + Ndy = 0$, where

$$M = (xysinxy + cos\, xy)y, N = (xysinxy - cosxy)x$$

and (1) is of the form $yf(xy)dx + xg(xy)dy = 0$

Now,

$$Mx - Ny = (xysinxy + cos\, xy)yx - (xysinxy - cosxy)xy$$
$$= 2xycos\, xy \neq 0$$

$$\therefore I.F. = \frac{1}{Mx - Ny} = \frac{1}{2xycosxy}.$$

Multiplying both sides of (1) by $I.F. = \frac{1}{2xycosxy}$, we get

$$\frac{1}{2}\left(ytanxy + \frac{1}{x}\right)dx + \frac{1}{2}\left(xtanxy - \frac{1}{y}\right)dy = 0$$

which is now exact.

Therefore, it's solution is

$$\frac{1}{2}\int\left(ytanxy+\frac{1}{x}\right)dx+\frac{1}{2}\int\left(-\frac{1}{y}\right)dy=c_1$$

$$\Rightarrow\frac{1}{2}(lnsec(xy)+lnx)-\frac{1}{2}lny=\frac{1}{2}lnc$$

$$\Rightarrow lnsec(xy)+ln\frac{x}{y}=lnc$$

$$\Rightarrow\frac{x}{y}sec(xy)=c$$

Example 2.

Solve $(x^3y^3+x^2y^2+xy+1)ydx+(x^3y^3-x^2y^2-xy+1)xdy$

Rule IV

If $\frac{1}{N}\left(\frac{\partial M}{\partial y}-\frac{\partial N}{\partial x}\right)$ is a function of x alone i.e. $f(x)$ then I.F. of $Mdx+Ndy=0$ is $I.F.=\ e^{\int f(x)dx}$.

Example 1. Solve $\left(y+\frac{y^3}{3}+\frac{x^2}{2}\right)dx+\frac{1}{4}(x\ +\ xy^2)dy=0$

Solution

The given equation is

$$\left(y+\frac{y^3}{3}+\frac{x^2}{2}\right)dx+\frac{1}{4}(x\ +\ xy^2)dy=0\quad\ldots\quad(1)$$

which is of the form $Mdx+Ndy=0$,

where $M=\left(y+\frac{y^3}{3}+\frac{x^2}{2}\right)$, $N=\frac{1}{4}(x+xy^2)$

$\therefore\ \frac{\partial M}{\partial y}=\frac{\partial}{\partial y}\left(y+\frac{y^3}{3}+\frac{x^2}{2}\right)=1+y^2$ and $\frac{\partial N}{\partial x}=\frac{1}{4}(1+y^2)$

Now,

$$\frac{1}{N}\left[\frac{\partial M}{\partial y}-\frac{\partial N}{\partial x}\right]=\frac{4}{x(1+y^2)}\left[1+y^2-\frac{1}{4}(1+y^2)\right]$$

$$=\frac{4}{x}-\frac{1}{x}=\frac{3}{x}=f(x)$$

Therefore,

$$I.F.=e^{\int f(x)dx}=e^{\int\frac{3}{x}dx}=e^{3lnx}=e^{lnx^3}=x^3.$$

Multiplying both sides log(1) by x^3, we get

$$\left(x^3y+\frac{x^3y^3}{3}+\frac{x^5}{2}\right)dx+\frac{1}{4}(x^4\ +\ x^4y^2)dy=0$$

(x^3y + **Fehler!**x^3y^3 + **Fehler!**x^5)dx + **Fehler!**(x^4 + x^4y^2)dy = 0,

Which is now exact. It's solution is

$$\int\left(x^3y+\frac{x^3y^3}{3}+\frac{x^5}{2}\right)dx+\int(terms\ of\ N\ free\ of\ x)dy=c$$

$$\Rightarrow\frac{x^4y}{4}+\frac{x^4y^3}{12}+\frac{x^6}{12}+\int 0dx=c_1$$

$$\Rightarrow\frac{x^4y}{4}+\frac{x^4y^3}{12}+\frac{x^6}{12}=c_1$$

$$\Rightarrow 3x^4y+x^4y^3+x^6=c$$

Hence, the required solution is

$$3x^4y+x^4y^3+x^6=c$$

where c is an arbitrary constant.

Rule V

If $\frac{1}{M}\left[\frac{\partial N}{\partial x}-\frac{\partial M}{\partial y}\right]$ is a function of y one i.e. $f(y)$, then I.F. of $Mdx+Ndy=0$ is $I.F.=e^{\int f(y)dy}$.

Example 1.

Solve $(2xy^4e^y+2xy^3+y)dx+(x^2y^4e^y-x^2y^2-3x)dy=0$

Solution

The given equation is

$$(2xy^4e^y+2xy^3+y)dx+(x^2y^4e^y-x^2y^2-3x)dy=0 \quad \ldots \quad (1)$$

which is of the form

$$Mdx+Ndy=0 \quad \ldots \quad (2)$$

where $M=(2xy^4e^y+2xy^3+y), N=(x^2y^4e^y-x^2y^2-3x)$.

$$\frac{\partial M}{\partial y}=8xy^3e^y+2xy^4e^y+6xy^2+1$$

And

$$\frac{\partial N}{\partial x}=2xy^4e^y-2xy^2-3$$

Now,

$$\frac{\partial N}{\partial x}-\frac{\partial M}{\partial y}=(2xy^4e^y-2xy^2-3)-(8xy^3e^y+2xy^4e^y+6xy^2+1)$$

$$=2xy^4e^y-2xy^2-3-8xy^3e^y-2xy^4e^y-6xy^2-1$$

$$=-8xy^3e^y-8xy^2-4=-4(2xy^3e^y+2xy^2+1)$$

$$\therefore \frac{1}{M}\left[\frac{\partial N}{\partial x}-\frac{\partial M}{\partial y}\right]=\frac{1}{(2xy^4e^y+2xy^3+y)}\times -4(2xy^3e^y+2xy^2+1)$$

$$=\frac{1}{y(2xy^3e^y+2xy^2+1)}\times -4(2xy^3e^y+2xy^2+1)$$

$$=-\frac{4}{y}=f(y)$$

$$\therefore I.F.=e^{\int f(y)dy}=e^{\int -\frac{4}{y}dy}=e^{-4lny}=e^{lny^{-4}}=e^{ln\frac{1}{y^4}}=\frac{1}{y^4}$$

Multiplying both sides of (1) by $I.F.=\frac{1}{y^4}$, we get

$$\frac{1}{y^4}(2xy^4e^y+2xy^3+y)dx+\frac{1}{y^4}(x^2y^4e^y-x^2y^2-3x)dy=0$$

$$or, \quad \left(2xe^y+\frac{2x}{y}+\frac{1}{y^3}\right)dx+\left(x^2e^y-\frac{x^2}{y^2}-\frac{3x}{y^4}\right)dy=0$$

which is now exact. It's solution is

$$\int\left(2xe^y+\frac{2x}{y}+\frac{1}{y^3}\right)dx+0=c$$

$$\Longrightarrow x^2e^y+\frac{x^2}{y}+\frac{x}{y^3}=c$$

Hence, the required solution of (1) is

$$x^2e^y+\frac{x^2}{y}+\frac{x}{y^3}=c$$

where c is an arbitrary constant.

Examples

Solve

$$(i)\ (x^2+y^2+1)dx+x(x-2y)dy=0$$

$$(ii)(y^4+2y)dx+(xy^3+2y^4-4x)dy=0$$